

Regression Week I: Foundations for linear models

Slide deck I: Course introduction and motivation

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What is the purpose of the regression section?

- ▶ To make you a useful statistician!
- ▶ Well, to really get you on your way
 - ▶ First year (STA1006S) is useful, but easily replicable by non-statisticians
 - ▶ STA2004F is (despite all the effort you put in) not immediately of great practical use
 - ▶ The practical skills and mathematical understanding you'll gain here are not easily replicable and can actually be used to assist decision-making

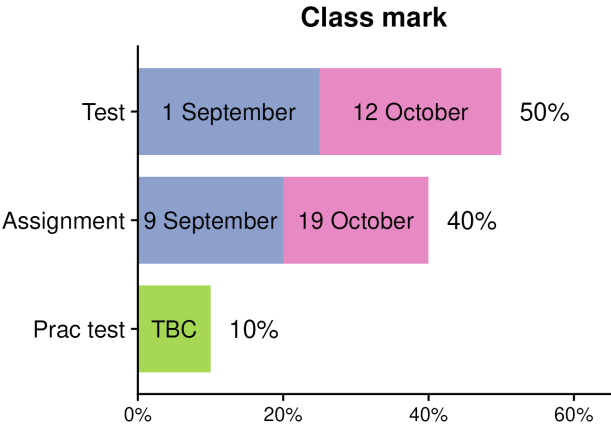
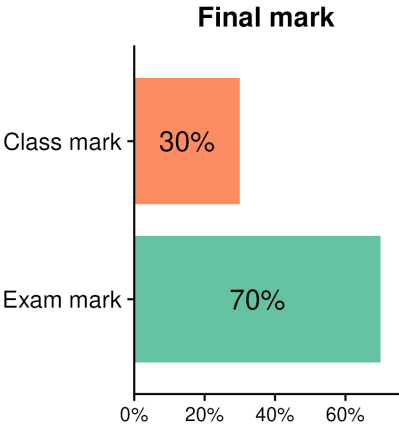
Key outcomes

- ▶ Mathematical understanding of simplest regression model
- ▶ Ability to code up an end-to-end modelling analysis
 - ▶ Reading in data
 - ▶ Analysing data
 - ▶ Displaying and interpreting results

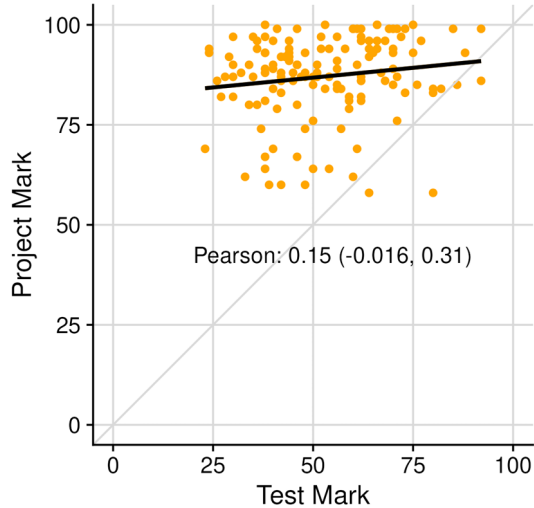
Lecture schedule

Weeks	Start date	End date	Section
1-6	27 Jul	4 Sep	Regression
Vac	7 Sep	11 Sep	
7-12	14 Sep	23 Oct	Experimental design

Assessment contributions



Relationship between assignment and test marks



Learning resources: content

- ▶ Regression notes:
 - ▶ Location: Module 1: Regression/Notes/NotesReg.pdf
 - ▶ Printed copies available at stats reception
- ▶ Slides:
 - ▶ Location: in Slides folder under each week's tab
- ▶ Lecture recordings:
 - ▶ Automatically uploaded to Lecture Videos/2026 Lecture Videos tab

Learning resources: testing I

- ▶ Q&A docs:
 - ▶ Comprehensive set of short questions on each slide deck's content
 - ▶ A review of all the facts, essentially
 - ▶ Will be in the Slides folder for each week's content
 - ▶ For example,
QARegNotesW1S0Prereq.pdf is in
Introduction/Slides folder
alongside
NotesRegW1S0Prereq.html
 - ▶ *Note:* these are not exam-style questions, but rather a way to a) remember the content, and b) check your understanding

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- On Amathuba (Module 1: Regression/Practice material/Tutorials)

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► Question bank:

- Appropriate questions over and above the tutorials
- Aim to progress from simpler to harder questions within each section
- On Amathuba (Module 1: Regression/Practice material/Question bank)
- Will be updated throughout the term

Advised learning/revision route for exam and test

- ▶ Attend lectures
- ▶ After each lecture, do relevant Q&A content
- ▶ Do tutorials (and attend them)
- ▶ Do question bank questions

Advice from past tutors

- ▶ Learning to code up analyses is perhaps the most useful part of the course.
- ▶ Don't underestimate the course - passing STA2004F is great, but this is a lot of content. Marks are often terrible.
- ▶ Be wary of the assignments - they count as much as the tests, and take a long time.
- ▶ The exam counts 70%, so don't presume that assignments and tests will carry you through. Focus on understanding the content.
- ▶ Questions to rehearse proofs are free marks.
- ▶ Experimental design in term four tends to be more difficult, so bank regression marks.

- ▶ Online lecture:
 - ▶ Day: Wednesday, 29 July
 - ▶ Time: 2pm
 - ▶ Link: Pending

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- ▶ Practicals:
 - ▶ Days: Wednesday, Thursday and Friday (29-31 July)
 - ▶ Times: 2-4pm
 - ▶ Category: R Intro Workshop

- ▶ In a computer lab, with a tutor, who will walk through coding up part of a regression analysis
- ▶ For example:
 - ▶ Fitting a regression model
 - ▶ Validating a regression model

Sign up for tutorials and practicals

- ▶ Frequency: weekly
- ▶ Start: week 2
- ▶ Signup:
 - ▶ Amathuba » Course info » Groups
 - ▶ Categories:
 - ▶ Practical: Computer Lab Session
 - ▶ Tutorial: STA2005S Tutorials

Office hours

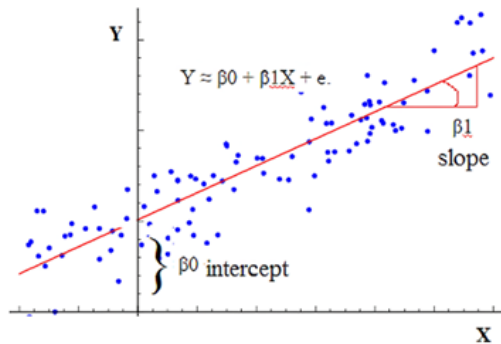
- ▶ Room: 5.52 PD Hahn (fifth floor, down the corridor from stats reception)
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- ▶ Purpose: any content-related questions
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- ▶ Purpose: any content-related questions
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- ▶ Email:
 - ▶ Address: `miguel.rod@uct.ac.za`
 - ▶ Purpose: any admin-related issues specific to the regression section

What is regression?

- ▶ $Y \sim f(X, \epsilon)$:
 - ▶ Y is the response variable (dependent variable)
 - ▶ X contains the explanatory variable(s) (or independent variables)
 - ▶ ϵ is the random variable
 - ▶ The function f describes how they relate
 - ▶ Y cannot be categorical
- ▶ A regression model describes how the distribution of a response Y depends on one or more explanatory variables X .
- ▶ For the linear regression models considered in this section, Y is continuous.
- ▶ Random variation is represented through an error term ϵ



Regression versus the correlation coefficient: round one

- ▶ Imagine we work out what the correlation is between advertising spending and revenue.
- ▶ In what way might the correlation coefficient be deficient?

Regression versus the correlation coefficient round two

- ▶ Now, suppose a doctor wants to decide whether a six-month old baby has a healthy weight.
- ▶ Why would the correlation coefficient not be useful?

Regression versus the t -test: round one

- ▶ Let's say you want to understand the relationship between blood pressure (the dependent variable) and taking a drug (the independent variable).
- ▶ Also assert that more males took the drug than females.
- ▶ Suppose we perform a t -test. How might someone object?

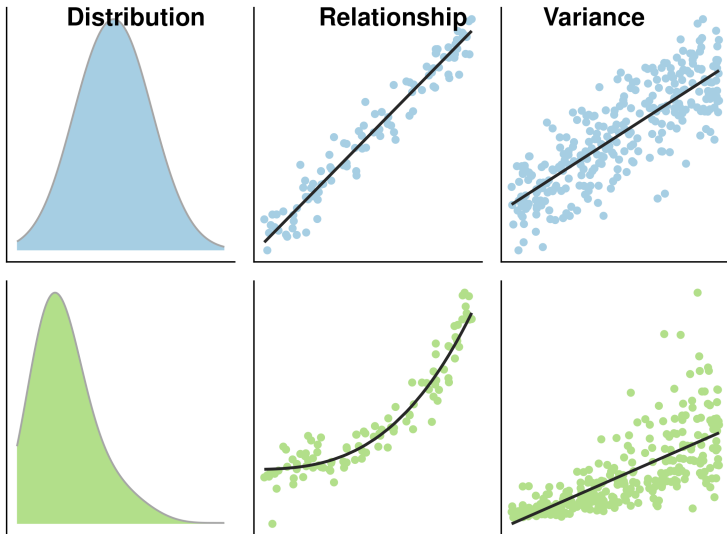
Regression versus the t -test: round two

- ▶ Suppose that we want to model blood pressure against BMI.
- ▶ What's the problem with the t -test in this case?

Toolbox



Assumptions

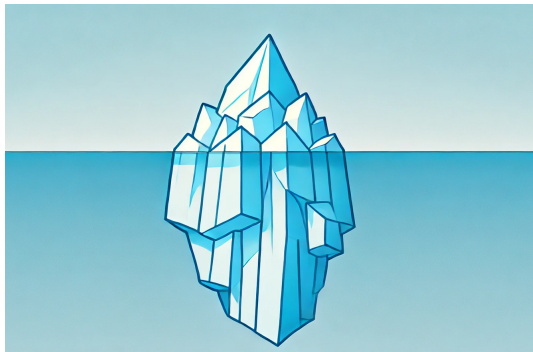


- ▶ In our course, we consider **linear regression** only:
 - ▶ Normally distributed response
 - ▶ Linear relationship with response variables
 - ▶ Constant variance

Regression becomes highly flexible

- ▶ Non-constant variance: mean-variance relationship, dependent on predictor variables, etc.
- ▶ Correlated errors:
 - ▶ within subsets of observations, e.g. measurements from the same individual, or
 - ▶ between “nearby” observations, e.g. time-series or spatial data
- ▶ Non-normally distributed data, e.g. count data
- ▶ Categorical dependent variables, e.g. disease status

Mathematical foundations to linear regression



Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.032594	0.457843	15.36	<2e-16 ***
TV	0.047537	0.002691	17.67	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.259 on 198 degrees of freedom

Multiple R-squared: 0.6119, Adjusted R-squared: 0.6099

F-statistic: 312.1 on 1 and 198 DF, p-value: < 2.2e-16

Regression section outline

- ▶ **Week one:** statistical and mathematical prerequisites (statistical distributions and linear algebra), and useful proofs regarding the multivariate normal distribution
- ▶ **Week two:** General linear model introduction (formulation, estimation and parameter distributions)
- ▶ **Week three:** General linear model inference (confidence intervals and hypothesis testing)
- ▶ **Week four:** ANOVA and model checking
- ▶ **Week five:** Variable transformations, variable selection and Gauss-Markov theorem
- ▶ **Week six:** PCA and bootstrapping

Prerequisites: Linear algebra and statistical theory I

Content, under Module 1:

Regression/Introduction/Prerequisites: Regression/Practice material:

- ▶ Linear algebra:

- ▶ Slides:

- SlidesRegW1S0PrereqLA.pdf

- ▶ Video:

- VideoRegW1S0PrereqLA.mp4

- ▶ Basics in R:

- DemRegW1S0LinearAlgebra.pdf

- ▶ Linear algebra + statistical

distributions:

- ▶ Linear algebra content from slides

- above, plus information on statistical distributions:

- NotesRegW1S0Prereq.html

- ▶ Q&A on notes:

- QARegNotesW1S0PrereqLA.pdf

Practice material, under Module 1:



Tutorial/TutReg1IntroMVNQuestions.

- ▶ The first few questions in this tutorial are about linear algebra